DTC P2263:00 [SKYACTIV-D 1.5]

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DTC P2263:00	Turbocharger solenoid valve range/performance
DETECTION CONDITION	• The following conditions are met, the difference between the target regulating valve opening angle and the actual regulating valve opening angle is **2 mm {0.08 in}** or more for a continuous 3 s: — Engine speed: above **700 rpm** — Turbocharger solenoid valve control duty value: **55 %** or more — Actuator position sensor: normal — HP-EGR control valve: Fully closed — Diesel particulate filter regeneration control is not performed • The following conditions are met, the difference between the target regulating valve opening angle and the actual regulating valve opening angle is **-6.2 mm {-0.25 in}** or less for a continuous 3 s: — Engine speed: above **700 rpm** — Turbocharger solenoid valve control duty value: **20 %** or less — Actuator position sensor: normal — HP-EGR control valve: Fully closed — Diesel particulate filter regeneration control is not performed **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM. • FREEZE FRAME DATA /Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Restricts upper limit of the engine speed. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the boost control. • PCM restricts engine-transaxle integration control. (ATX)
POSSIBLE CAUSE	• Vacuum piping malfunction — Power brake unit—Vacuum pump — Vacuum pump—Check valve — Check valve—Vacuum chamber — Vacuum chamber—Turbocharger solenoid valve — Turbocharger solenoid valve—Turbocharger actuator • Vacuum chamber (built-into cylinder head cover) malfunction • Turbocharger solenoid valve malfunction • Check valve malfunction • Actuator position sensor (turbocharger with variable turbine geometry) malfunction • Turbocharger with variable turbine geometry malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT VACUUM PIPING** • Inspect vacuum piping at the following: (See INTAKE-AIR SYSTEM VACUUM HOSE ROUTING DIAGRAM [SKYACTIV-D 1.5].) (See CONVENTIONAL BRAKE SYSTEM LOCATION INDEX.) — Power brake unit—Vacuum pump — Vacuum pump—Check valve — Check valve—Vacuum chamber — Vacuum chamber—Turbocharger solenoid valve — Turbocharger solenoid valve—Turbocharger actuator • Is there hose leakage or damage in the vacuum piping?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 10.
		No	Go to the next step.
5	**INSPECT VACUUM CHAMBER** • Inspect the vacuum chamber. (See VACUUM CHAMBER INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the cylinder head cover, then go to Step 10. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**INSPECT TURBOCHARGER SOLENOID VALVE** • Inspect the turbocharger solenoid valve. (See TURBOCHARGER SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger solenoid valve, then go to Step 10. (See TURBOCHARGER SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**INSPECT CHECK VALVE** • Inspect the check valve. (See CHECK VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the check valve, then go to Step 10. (See CHECK VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT ACTUATOR POSITION SENSOR (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY)** • Inspect the actuator position sensor (turbocharger with variable turbine geometry). (See ACTUATOR POSITION SENSOR (TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY) INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger with variable turbine geometry, then go to Step 10. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**INSPECT TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY** • Inspect the turbocharger with variable turbine geometry. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger with variable turbine geometry, then go to the next step. (See TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. • Drive the vehicle for **1 min**. at a speed of **35—50 km/h {22—31 mph}**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.